County matching: Glaeser and Gottlieb (2009)

Control: Match of Historical to Modern Geographic Units — Glaeser and Gottlieb (2009). At State level, this is a historical-instrument choice.

1. Match county identities. This option matches historical county identities to the project's current county units. The matching uses county codes and explicit corrections for renamed, recoded or otherwise identified counties. Current county membership then determines the destination MSA or state. This note describes the Glaeser–Gottlieb-style county matching implemented in the project.

2. Add whole historical populations. Let P_{jt} be the population of historical county j in census year t, and let $m_{jg,t} = 1$ when that historical county is assigned to modern region g. Otherwise, $m_{jg,t} = 0$. The matched population is

$$P_{g,t}^{\text{match}} = \sum_{j} m_{jg,t} P_{jt}.$$

Under Levels, the instrument is $z_{g,t} = P_{g,t}^{\text{match}}/1{,}000$. It measures thousands of people. Natural log instead uses $\log P_{g,t}^{\text{match}}$ for positive totals. The population total is used with both the employment and employment-density approaches. The minimum-overlap control does not apply to this method.

3. Matching does not divide population by shared area. If a historical county later splits, the match can assign its whole population to one identified successor. It does not distribute that population among all modern descendants. If all descendants remain in the same MSA, the MSA sum can nevertheless be unchanged. If a split crosses the modern MSA's boundary, matching and area allocation can produce different totals. The county identities, rather than the proportion of land inside the modern region, determine the assignment.

4. Unresolved sources remain a limitation. An uncertain identity is not assigned arbitrarily. Missing populations for matched counties and known source gaps make affected totals unavailable. Jointly reported populations require special handling: an ambiguous county allocation cannot be recovered merely by matching names. A known joint total can be used at State level when all relevant counties belong to that state. Consequently, available totals and samples can differ from the area-weighted method. Comparing the two on the same estimation sample helps distinguish the effect of population assignment from the effect of including different observations.